BIKE RENTAL MANAGEMENT SYSTEM

Project Synopsis Submitted
In Partial Fulfillment of the Requirements
for Degree of Bachelor of Technology

in

COMPUTER SCIENCE

by

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September, 2025

ABSTRACT

This project aims to develop a **web-based Bike Rental Management System** that enables customers to rent bikes for personal use without the involvement of a rider. Traditional bike rental shops still rely on manual paperwork, leading to inefficiency, errors, and lack of transparency. Our system addresses this by providing a **digital platform** for users to register, verify their identity, check bike availability in real-time, book a vehicle, and make secure payments.

The proposed system will include an **admin module** to manage bikes, track bookings, monitor returns, and generate reports. Technologies such as **HTML, CSS, Bootstrap, JavaScript (frontend), PHP (backend), and MySQL (database)** will be used.

Expected outcomes include **streamlined rental operations, reduced manual errors, improved customer convenience, and better analytics** for shop owners. The project not only digitizes bike rental shops but also contributes towards sustainable mobility by promoting bike usage.

INTRODUCTION

Bike rentals are a growing alternative to traditional transport, especially for daily commuters and tourists. However, many shops still use **manual registers and cash-based transactions**, causing inefficiency. The increasing demand for flexible, affordable, and eco-friendly transport highlights the importance of developing an automated rental system.

The project’s scope is to design a **centralized, user-friendly system** that handles the entire rental process — from registration and booking to billing and return. It will be scalable for both **small local shops** and **large rental companies**.

PROBLEM STATEMENT

The existing manual rental process is **time-consuming, prone to errors, lacks transparency, and offers no real-time bike availability updates**. Customers face inconvenience, while shopkeepers struggle with record-keeping, fraud prevention, and reporting. There is no integrated system to handle **user authentication, booking management, and digital payments**.

OBJECTIVE

- To design and develop a **responsive web application** for bike rentals.
- To allow customers to **register, authenticate, and book bikes online**.
- To integrate a **secure payment system** and automated billing.
- To build an **admin panel** for managing bikes, customers, and rentals.
- To ensure **real-time availability** updates and maintain rental history.

LITERATURE REVIEW

Existing platforms such as Ola, Uber Moto, and Rapido provide **bike taxi services** but face legal restrictions in many states, including Maharashtra. Self-drive bike rental platforms like **Bounce, Yulu, and Royal Brothers** exist but are **limited to metro cities** and often require mobile apps.

Traditional local shops rely on **manual registers and ID verification**, which is inefficient. Our project fills the gap by offering a **web-based, shop-friendly solution** that digitizes local rentals and makes them more accessible.

PROPOSED SYSTEM

The proposed Bike Rental Management System will:

- Provide **customer features**: registration, login, ID upload, booking, payment, return confirmation.
- Provide **admin features**: add/update/delete bikes, track rentals, view reports.
- Be built using **PHP + MySQL (backend), HTML/CSS/Bootstrap/JavaScript (frontend)**.
- Include a **system architecture** with three modules: Customer, Admin, Database.

Advantages over existing system:

- Real-time bike availability.
- Automated record keeping.
- Secure and transparent transactions.
- Scalable for small and large businesses

METHODOLOGY

- **Step 1:** Requirement analysis & system design.
- **Step 2:** Database design (ER diagram, schema).
- **Step 3:** Frontend development (HTML, CSS, Bootstrap, JS).
- **Step 4:** Backend development (PHP + MySQL).
- **Step 5:** Integration of modules & payment system.
- **Step 6:** Testing (unit, integration, user acceptance).
- **Step 7:** Deployment & demonstration.

HARDWARE AND SOFTWARE REQUIREMENTS

Hardware:

- Processor: Intel i3 or higher
- RAM: 4GB+
- Storage: 100GB+

Software:

- OS: Windows/Linux
- IDE: VS Code / Eclipse
- Database: MySQL
- Languages: HTML, CSS, Bootstrap, JavaScript, PHP
- Web Server: Apache

RESULTS

- **Customer Portal:** Login, bike availability, booking, invoice.
- **Admin Portal:** Manage bikes, customers, bookings, reports.
- **Outputs:** Reports on rental usage, earnings, customer activity.
- **Screenshots:** Login page, booking form, admin dashboard.

APPLICATIONS

- Local bike rental shops.
- Tourism hotspots (Goa, Manali, Ladakh).
- Corporate campuses offering employee transport.
- College/university campuses for student mobility.

FUTURE ENHANCEMENTS

- Mobile app integration (Android/iOS).
- GPS tracking of bikes.
- Subscription-based plans for daily commuters.
- AI-based demand forecasting and smart pricing

REFERENCES

[1] GitHub having Bike rental system projects: sabbir-hossain-abir/Bike-Rental-System

[2] Websites having Bike Rental Service: https://bikerentalmanager.com/